Poincaré Duality

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of poincaré duality, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AT-05 / Algebraic Topology / Report v1

I. TARGET AND IMPORT CONTRACT

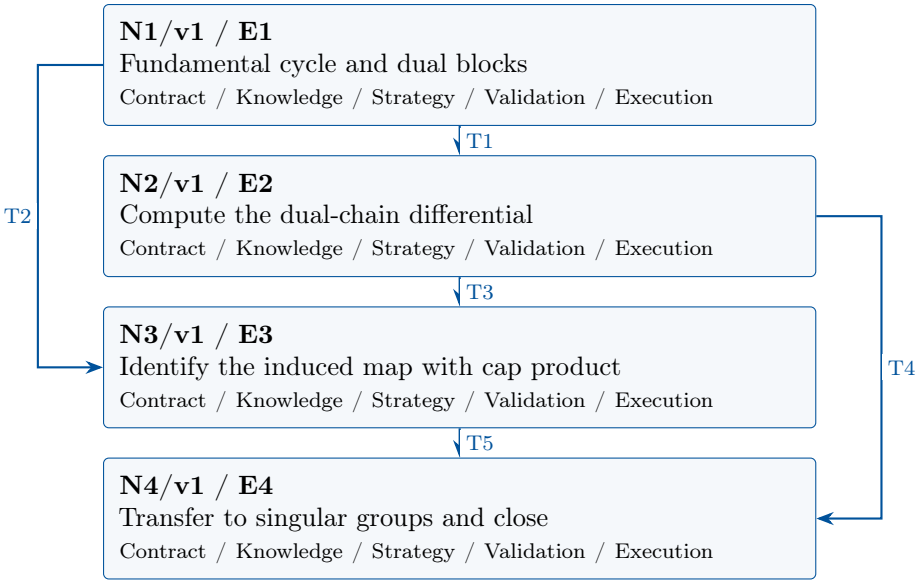
Theorem I.1. *For a closed connected oriented smooth n-manifold M, prove cap product with its integral fundamental class gives an isomorphism $H^k(M;\mathbb{Z}) \to H_{n-k}(M;\mathbb{Z})$ for $0 \le k \le n$.*

A. Foundations and permitted imports

Use a finite smooth triangulationK of the closed oriented manifold, compatible orientations on top simplices, and its barycentric subdivision. Import simplicial/cellular-to-singular homology and cohomology comparison and subdivision invariance. In a triangulated manifold the closed star of each simplex is contractible, and barycentric dual blocks form a cell decomposition: the link is a sphere and coning its subdivided chains gives the complementary-dimensional ball. Use the chain definition of cap product with identity $\partial([M] \text{ cap } \alpha) = (-1)^{(k+1)}[M] \text{ cap } \delta\alpha$ for a closed fundamental cycle and a degree-k cochain, with a fixed matching cap convention. No duality is imported; the dual differential and identification are established below.

Use a compatible smooth (hence combinatorial) triangulation, so links have the required sphere structure. The singular-chain filling in a closed star is provided by a contraction of that star, not by a claim that an arbitrary contractible space is literally a geometric cone.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The fundamental cycle and a basis pairing between simplicial cochains and dual cellular chains.

Receiving activity. N2: Compute the dual-chain differential

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Oriented dual blocks and incidence $numbers[\tau : \sigma]$; prior exports E1/v1

T2: N1/E1/v1 to N3. The fundamental cycle and a basis pairing between simplicial cochains and dual cellular chains.

Receiving activity. N3: Identify the induced map with cap product

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Dual map D and cap map againstc; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. A degree-reversing chain identification inducing $H^k(K;Z) \cong H_{n-k}(dualK;\mathrm{Z})$.

Receiving activity. N3: Identify the induced map with cap product

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Dual map D and cap map againstc; prior exports E1/v1, E2/v1

T4: N2/E2/v1 to N4. A degree-reversing chain identification inducing $H^k(K;Z) \cong H_{n-k}(dualK;\mathrm{Z})$.

Receiving activity. N4: Transfer to singular groups and close

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Chain identification and cap comparison; prior exports E2/v1, E3/v1

T5: N3/E3/v1 to N4. The dual-cell isomorphism is the cap-product map $[\mathrm{M}]\ cap(-)$.

Receiving activity. N4: Transfer to singular groups and close

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Chain identification and cap comparison; prior exports E2/v1, E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Fundamental cycle and dual blocks

Object and role. Pair an oriented simplex with its transverse dual block.

Inputs. Oriented triangulation of a closedn-manifold; foundational imports stated in §1

C — Contract

Own E1/v1: The fundamental cycle and a basis pairing between simplicial cochains and dual cellular chains.

K — Knowledge

Retain Oriented triangulation of a closedn-manifold; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Pair an oriented simplex with its transverse dual block.

V — Validation

Closedness cancels all faces; orientability makes their signs globally compatible.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Letc be the sum of compatibly orientedn-simplices.

N1.2 Every $(n-1)$-face is shared twice with opposite induced signs, so $\partial c = 0$.

N1.3 Its local coefficient at every oriented manifold point is$+1$, so it represents the integral fundamental class[M].

N1.4 For an orientedk-simplex σ, its dual blockD σ is the union of barycentric flags beginning at σ and continuing through cofaces up to top-dimensional simplices.

N1.5 It has dimension $n-k$.

N1.6 OrientD σ so the ordered orientation of σ followed by its transverse dual orientation gives the orientation of M.

N1.7 These blocks bijectively pair k-simplices with $(n-k)$-cells.

Export E1/v1. The fundamental cycle and a basis pairing between simplicial cochains and dual cellular chains.

Acceptance gate. Closedness cancels all faces; orientability makes their signs globally compatible. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Closedness cancels all faces; orientability makes their signs globally compatible. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Compute the dual-chain differential

Object and role. Compare boundary flags for σ and its immediate cofaces.

Inputs. Oriented dual blocks and incidence $numbers[\tau : \sigma]$; prior exports E1/v1

C — Contract

Own E2/v1: A degree-reversing chain identification inducing $H^k(K;Z) \cong H_{n-k}(dualK;\mathrm{Z})$.

K — Knowledge

Retain Oriented dual blocks and incidence $numbers[\tau : \sigma]$; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Compare boundary flags for σ and its immediate cofaces.

V — Validation

Ignoring signs would only justify a mod 2 statement, not integral duality.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 The codimension-one boundary faces of $D\sigma$ correspond to (k+1)-cofaces τ of σ and are copies of $D\tau$.

N2.2 Other flag faces occur twice with opposite subdivision signs and cancel.

N2.3 $If[\tau : \sigma]$ is the signed simplex incidence, moving the normal-to-σ direction across itsk tangent directions and using outward-first boundary orientation gives the coefficient $(-1)^{(k+1)}[\tau : \sigma]$.

N2.4 Thus $\partial D\sigma = (-1)^{(k+1)} \sum_{\tau \supset \sigma} [\tau : \sigma] D\tau$.

N2.5 For the cochain basis $\sigma*$, $\delta\sigma* = \sum [\tau : \sigma]\tau*$.

N2.6 Hence the degree-reversing basis map D satisfies $\partial D\alpha = (-1)^{(k+1)} D\delta\alpha$.

N2.7 It sends cocycles to cycles and coboundaries to boundaries, with an inverse on bases and the same differential relation.

Export E2/v1. A degree-reversing chain identification inducing $H^k(K;Z) \cong H_{n-k}(dualK;\mathrm{Z})$.

Acceptance gate. Ignoring signs would only justify a mod 2 statement, not integral duality. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Ignoring signs would only justify a mod 2 statement, not integral duality. Recheck dependent nodes N3, N4.

C. N3/v1: Identify the induced map with cap product

Object and role. Use local carriers to compare two chain constructions, not only abstract ranks.

Inputs. Dual map D and cap map againstc; prior exports E1/v1, E2/v1

C — Contract

Identify cap product with the dual-cell isomorphism.

K — Knowledge

Import E1/v1, E2/v1, the cap-chain formula, and contractible closed stars.

S — Strategy

Use local carriers to compare two chain constructions, not only abstract ranks.

V — Validation

An abstract group isomorphism would not identify the specified canonical map.

E — Execution

Build the carrier homotopy and publish E3/v1.

Numbered mathematical activities

N3.1 View the reversed cochain complex as a chain complex with degree $r = n - k$ and differential $(-1)^{(k+1)}\delta$.

N3.2 After mapping dual cellular chains to singular chains in the subdivision, both D and $C : \alpha \mapsto c \text{ cap } \alpha$ are chain maps to singular chains.

N3.3 For a basis $\sigma*$ both images are supported in the closed star of σ: dual flags lie there, and every cap term comes from a top simplex containing σ.

N3.4 The carrier is nested under the differential since cofaces have stars contained in that of σ.

N3.5 In degree 0, for a top-simplex cochain, the dual barycenter and cap-selected vertex have the same oriented augmentation.

N3.6 Construct a chain homotopy upward on this free basis: if H is already defined one degree lower, $z = (D - C)(\sigma*) - H(d\sigma*)$ is a cycle in star σ, with zero augmentation when it is a0-cycle.

N3.7 The star is contractible, so this cycle bounds; use the induced singular-chain contraction to choose a filling $H(\sigma^*)$.

N3.8 Inductively $D - C = \partial H + Hd$.

N3.9 Thus the two maps induce the same map on homology.

N3.10 This is the explicit local-carrier comparison between dual blocks and the cap construction.

Export E3/v1. The dual-cell isomorphism is the cap-product map [M] $cap(-)$.

Acceptance gate. Identify the canonical cap-product map, not merely isomorphic groups.

Targeted repair. Rebuild the carrier homotopy, then recheck N4.

D. N4/v1: Transfer to singular groups and close

Object and role. Apply the imported comparison maps compatibly.

Inputs. Chain identification and cap comparison; prior exports E2/v1, E3/v1

C — Contract

Own E4/v1: Integral Poincaré duality with the specified fundamental-class cap map.

K — Knowledge

Retain Chain identification and cap comparison; prior exports E2/v1, E3/v1; keep their assumptions and versions attached.

S — Strategy

Apply the imported comparison maps compatibly.

V — Validation

The theorem is for closed oriented manifolds; boundary and nonorientability change the target coefficients or relative groups.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Simplicial cohomology of K and cellular homology of its dual decomposition identify with the respective singular groups of M.

N4.2 These comparison maps commute with subdivision and cap products by the chain constructions used in node3.

N4.3 Since D is an isomorphism on the reversed complexes, it is an isomorphism on their homology.

N4.4 Since C induces the same map, cap product with[M] gives $H^k(M;Z) \to H_{n-k}(M;\mathrm{Z})$ as an isomorphism for every $0 \le k \le n$.

N4.5 Reversing orientation negates[M] and the resulting map but preserves its invertibility.

Export E4/v1. Integral Poincaré duality with the specified fundamental-class cap map.

Acceptance gate. The theorem is for closed oriented manifolds; boundary and nonorientability change the target coefficients or relative groups. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: The theorem is for closed oriented manifolds; boundary and nonorientability change the target coefficients or relative groups. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Fundamental cycle and dual blocks

1. Letc be the sum of compatibly orientedn-simplices.
2. Every $(n-1)$-face is shared twice with opposite induced signs, so $\partial c = 0$.
3. Its local coefficient at every oriented manifold point is$+1$, so it represents the integral fundamental class[M].
4. For an orientedk-simplex σ, its dual blockD σ is the union of barycentric flags beginning at σ and continuing through cofaces up to top-dimensional simplices.
5. It has dimension $n-k$.
6. OrientD σ so the ordered orientation of σ followed by its transverse dual orientation gives the orientation of M.
7. These blocks bijectively pair k-simplices with $(n-k)$-cells.

B. §4.N2 — Compute the dual-chain differential

1. The codimension-one boundary faces of $D\sigma$ correspond to (k$+1$)-cofaces τ of σ and are copies of $D\tau$.
2. Other flag faces occur twice with opposite subdivision signs and cancel.
3. $If[\tau:\sigma]$ is the signed simplex incidence, moving the normal-to-σ direction across itsk tangent directions and using outward-first boundary orientation gives the coefficient $(-1)^{(k+1)}[\tau:\sigma]$.
4. Thus $\partial D\sigma = (-1)^{(k+1)} \sum_{\tau\supset\sigma}[\tau:\sigma]D\tau$.
5. For the cochain basis $\sigma*$, $\delta\sigma* = \sum[\tau:\sigma]\tau*$.
6. Hence the degree-reversing basis map D satisfies $\partial D\alpha = (-1)^{(k+1)}D\delta\alpha$.
7. It sends cocycles to cycles and coboundaries to boundaries, with an inverse on bases and the same differential relation.

C. §4.N3 — Identify the induced map with cap product

1. View the reversed cochain complex as a chain complex with degree $r = n-k$ and differential $(-1)^{(k+1)}\delta$.

2. After mapping dual cellular chains to singular chains in the subdivision, both D and $C : \alpha \mapsto c$ cap α are chain maps to singular chains.

3. For a basis $\sigma*$ both images are supported in the closed star of σ: dual flags lie there, and every cap term comes from a top simplex containing σ.

4. The carrier is nested under the differential since cofaces have stars contained in that of σ.

5. In degree 0, for a top-simplex cochain, the dual barycenter and cap-selected vertex have the same oriented augmentation.

6. Construct a chain homotopy upward on this free basis: if H is already defined one degree lower, $z = (D - C)(\sigma*) - H(d\sigma*)$ is a cycle in star σ, with zero augmentation when it is a0-cycle.

7. The star is contractible, so this cycle bounds; use the induced singular-chain contraction to choose a filling $H(\sigma^*)$.

8. Inductively $D - C = \partial H + Hd$.

9. Thus the two maps induce the same map on homology.

10. This is the explicit local-carrier comparison between dual blocks and the cap construction.

D. §4.N4 — Transfer to singular groups and close

1. Simplicial cohomology of K and cellular homology of its dual decomposition identify with the respective singular groups of M.

2. These comparison maps commute with subdivision and cap products by the chain constructions used in node3.

3. Since D is an isomorphism on the reversed complexes, it is an isomorphism on their homology.

4. Since C induces the same map, cap product with[M] gives $H^k(M;Z) \to H_{n-k}(M;\mathrm{Z})$ as an isomorphism for every $0 \le k \le n$.

5. Reversing orientation negates[M] and the resulting map but preserves its invertibility.

□

V. WORKED ORCHESTRATION

On an oriented torus, $H^0 = Z$ pairs with the fundamental2-cycle, $H^2 = Z$ pairs with a point class, and $H^1 = Z^2$ pairs with the two complementary1-cycle directions. Using

product orientation, the degree 1 classes are exchanged with complementary cycles up to the cap convention's signs. The dual-block proof supplies the isomorphism rather than only comparing these ranks.

A. Boundary and hypothesis audit

For a disk, $H^0(D^2;Z) = Z$ but $H_2(D^2;Z) = 0$, so the absolute closed-manifold formula fails. For a nonorientable closed manifold an untwisted integral fundamental class need not exist. Validation blocks the initial fundamental-cycle gate; relative or local-coefficient duality is a distinct theorem.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For a disk, $H^0(D^2;Z) = Z$ but $H_2(D^2;Z) = 0$, so the absolute closed-manifold formula fails. For a nonorientable closed manifold an untwisted integral fundamental class need not

exist. Validation blocks the initial fundamental-cycle gate; relative or local-coefficient duality is a distinct theorem.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Allen Hatcher, Algebraic Topology: covering spaces, cellular homology, and cap products.